

Case Study: Team Performance Analysis (Olympic Machine Learning)

Student: R Virshin | **Roll No:** 150096724147 | **Course:** Machine Learning | **Date:** October 2026
Live Deployed App: <https://virshin1-team-performance-analysis-app-kwww1u.streamlit.app/> | **GitHub:** <https://github.com/Virshin1/Team-Performance-Analysis>

Executive Summary & Deliverable Synthesis: This report presents the formal machine learning investigation for the assigned coursework problem: 'A team wants to understand which measurable factors are associated with stronger team results (With Proper Justification).' We construct an empirical delegation-level dataset of **2,264 team-year observations** across **223 National Olympic Committees (NOCs)** spanning 15 Summer Olympic Games (1960–2016). We establish a dual-objective machine learning system: a **Random Forest Classifier** for predicting podium attainment (Accuracy: **90.51%**, F1: **0.8782**, ROC-AUC: **0.9584**) and a **Gradient Boosting Regressor** for projecting exact medal counts (Normal Train R2: **0.9863**, Normal Test R2: **0.8838**, MAE: **1.80** medals). Feature importance and econometric significance testing confirm that **78.9%** of competitive success is governed by organizational roster scale and discipline breadth, **11.3%** by institutional quadrennial momentum, and **9.8%** by delegation physical biometrics. A complete wide-format decision support application is deployed on Streamlit Cloud.

1. Formal Problem Definition & Formulation

Assigned Problem Statement: A team wants to understand which measurable factors are associated with stronger team results (With Proper Justification).

Real-World Context: National sports ministries and Olympic committees operate under strict budget allocations. Athletic directors face strategic trade-offs: should capital be allocated toward intensive physiological/biometric conditioning, broadening qualification quotas across diverse sports disciplines, or maintaining elite pipelines in legacy sports? This project converts this operational challenge into two concrete mathematical tasks: **Task A (Binary Classification)** to predict whether a national team wins at least one medal ($has_podium \in \{0, 1\}$), and **Task B (Continuous Estimation)** to forecast total medal volume ($total_medals \in [0, 121]$).

2. Dataset Architecture, Provenance & Quality Observations

Data Source: The dataset was constructed from historical Olympic athlete archives encompassing over 200,000 competitor-event entries spanning the modern Summer Olympic era (1960 Rome to 2016 Rio de Janeiro).

Critical Quality Rule (Event-Level Deduplication): In squad events (e.g. Football 18 players, Rowing 8 crew), every participant is awarded a physical medal. Treating raw athlete rows as individual medals inflates team-sport medals by 10x–18x. Records were deduplicated strictly to unique (*Games, Year, Event, Medal*) tuples before delegation aggregation.

Biometric Imputation: Missing athlete anthropometry for early decades was imputed using *median grouped by sport discipline and sex*, avoiding distortive global mean shifts (e.g. jockeys vs shot-putters).

Lagged Momentum: *prev_medals* was engineered using quadrennial historical lag per nation to measure institutional sports continuity.

Feature / Dimension	Type	Observed Range	Operational Definition & Role
total_athletes	Integer	1 to 650+	Unique competitors on official national roster (Primary scale foundation)
num_events	Integer	1 to 250+	Distinct competition categories contested (Medal opportunity breadth)
total_entries	Integer	1 to 800+	Total athlete-event registrations (Workload and dual entrants)
num_sports	Integer	1 to 32	Distinct sports disciplines contested (Hedges against sport shocks)
prev_medals	Float	0 to 120+	Medals secured in preceding Olympiad (Institutional funding momentum)
mean_age	Float	18.0 to 35.0	Average competitor age in years (Maturity vs youth athleticism)
female_ratio	Float	0.00 to 1.00	Proportion of female competitors (Gender parity and program breadth)
mean_height / weight	Float	155-195cm / 45-105kg	Average physical dimensions across delegation disciplines
mean_bmi	Float	18.0 to 28.0	Computed Body Mass Index (weight / height ²)
is_host	Binary	0 or 1	Host country status (Automatic qualification quota advantages)

3. Exploratory Data Analysis & Empirical Observations

Pareto Skewness: Olympic medal outcomes follow an extreme power-law distribution. Of 2,264 delegations, **59.94% (1,357)** won zero medals, while the top 5 nations historically capture >40% of all medals. The median roster of medal-winning nations is **142 athletes across 16 sports**, compared to **11 athletes across 4 sports** for non-winning nations (a 12.9x scale contrast).

Linear Correlation Rankings:

- *For Total Medals Volume:* **prev_medals (r = +0.901)**, total_entries (r = +0.834), total_athletes (r = +0.834), num_events (r = +0.777), num_sports (r = +0.622).
- *For Winning Any Podium Medal:* **num_sports (r = +0.702)**, num_events (r = +0.635), total_athletes (r = +0.575), total_entries (r = +0.572), prev_medals (r = +0.370).
- *Biometrics:* Physical dimensions (mean_age r=+0.012, mean_bmi r=-0.006) exhibit near-zero linear correlation because elite physical demands cancel out across disparate sports.

4. Machine Learning Methodology: Technical Justification for Random Forest

Random Forest was chosen as the singular champion architecture over linear models, support vector machines, and neural networks based on 5 technical justifications:

- **1. Intrinsic Impurity Attribution (Gini MDI):** Directly solves the project mandate ('understand which measurable factors are associated with stronger results') by quantifying the exact predictive contribution of each feature across 200 de-correlated estimators.
- **2. Non-Linear Interaction Detection:** Olympic medal returns are non-linear; expanding roster size within a single sport yields diminishing marginal returns, but expanding into an entirely new sport adds independent medal probabilities. Tree splits naturally capture these multi-variable interaction thresholds without artificial polynomial expansions.
- **3. Outlier Invariance:** Powerhouse delegations (e.g. USA with 600+ athletes and 100+ medals) act as extreme leverage points in Euclidean and regression models. Decision trees split on rank ordering rather than geometric distance, insulating predictions from outlier distortion.
- **4. Bagging Variance Reduction:** Single decision trees overfit historical tournaments. Ensembling 200 trees trained on bootstrap samples with random subspace feature selection (sqrt(p)) decorrelates trees and minimizes generalization variance.
- **5. Monotonic & Scale Invariance:** Seamlessly processes continuous biometrics (BMI), bounded ratios (female_ratio), and large unbounded counts (total_athletes) without distortive mathematical transformations.

5. Model Evaluation, Benchmarks & Normal R2 Results

The models were evaluated on the held-out test partition (\$n = 453\$, 20% stratified split). As instructed, regression performance is evaluated using **Normal R2** (Train R2 and Test R2) rather than cross-validation averages.

Task & Architecture	Evaluation Metric	Train Score	Test Score	Baseline Benchmark Comparison
Random Forest (Classifier)	Classification Accuracy	99.12%	90.51%	Dummy Majority Baseline: 59.94% (+30.57% lift)
Random Forest (Classifier)	Precision (Medal Winners)	98.90%	90.12%	Dummy Baseline: 0.00% (High confidence podium calls)
Random Forest (Classifier)	Recall (Medal Winners)	98.89%	85.64%	Captures 155 of 181 true medal-winning nations
Random Forest (Classifier)	F1-Score	0.9889	0.8782	Harmonic balance of precision and recall
Random Forest (Classifier)	ROC-AUC Score	0.9995	0.9584	Chance Level: 0.5000 (Exceptional threshold separation)
Gradient Boosting (Regressor)	Normal R2	0.9863	0.8838	Explains 88.4% of total medal variance on unseen data
Gradient Boosting (Regressor)	Mean Absolute Error (MAE)	0.91 medals	1.80 medals	Dummy Baseline: 7.74 medals (76.7% error reduction)
Gradient Boosting (Regressor)	Root Mean Squared Error (RMSE)	1.75 medals	5.01 medals	Dummy Baseline: 14.69 medals (65.9% error reduction)

6. Statistical Justification & Feature Importance Hierarchy

Direct Answer to the Problem Statement: To identify which measurable factors associate with stronger results with proper justification, Random Forest Gini MDI importance and econometric OLS hypothesis testing group the 11 variables into three distinct structural tiers:

Ra nk	Factor Name	Category	MDI Imp.	Corr (Medals)	Corr (Podium)	Statistical Significance & Strategic Finding
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1	total_athletes	Scale	25.53%	+0.834	+0.575	$p < 0.0001$ (beta = +4.82). Primary qualification foundation.
2	num_events	Scale	25.00%	+0.777	+0.635	$p < 0.0001$ (beta = +2.32). Measures medal opportunity breadth.
3	total_entries	Workload	16.47%	+0.834	+0.572	$p < 0.0001$. Athlete multi-event workload and specialization.
4	num_sports	Breadth	11.92%	+0.622	+0.702	$p = 0.0017$. Highest predictor of securing at least ONE medal.
5	prev_medals	Momentum	11.27%	+0.901	+0.370	$p < 0.0001$. Captures multi-cycle institutional funding stability.
6	mean_age	Biometric	2.83%	+0.012	+0.074	$p = 0.412$. Maturity vs physical peak (sport-dependent).
7	female_ratio	Demographic	1.89%	+0.159	+0.119	$p = 0.0384$ (beta = +0.89). Significant parity advantage.
8	mean_height	Biometric	1.78%	+0.110	+0.196	Discipline-specific advantage cancels out in aggregate.
9	mean_bmi	Biometric	1.74%	-0.006	+0.0001	$p = 0.891$. No linear aggregate relationship to medals.
10	mean_weight	Biometric	1.57%	+0.071	+0.128	Discipline-specific advantage cancels out in aggregate.
11	is_host	Contextual	< 0.01%	+0.304	+0.100	$p = 0.0007$ (beta = +4.12 medals). Host quota entry advantage.

7. Strategic Recommendations & Examination Defense Evidence

Policy Recommendation 1 (Qualification Breadth): Delegations with <15 athletes win medals <5% of the time, whereas expanding to >45 athletes elevates podium probability past 65%. Sports federations should prioritize multi-discipline Olympic qualifying pathways over extreme specialization.

Policy Recommendation 2 (Funding Quadrennials): Because *prev_medals* exhibits an $r = 0.901$ correlation with current output, cutting sports budgets following a single disappointing Olympiad induces compounding multi-cycle decline.

Viva Voce Defense Evidence:

- *Why Random Forest?* It resolves non-linear interaction thresholds and Pareto skewness without overfitting, while natively yielding Gini impurity factor attributions.
- *Why Normal R2?* The project strictly reports standard out-of-sample Normal Train R2 (0.9863) and Test R2 (0.8838) to demonstrate generalization without cross-validation fold averaging.

8. Interactive Streamlit Prototype & Cloud Deployment Verification

The finalized machine learning system is deployed live on Streamlit Community Cloud:

<https://virshin1-team-performance-analysis-app-kwww1u.streamlit.app/>.

The interface features a responsive wide layout, 1-click preset benchmark scenarios (Global Powerhouse, Mid-Tier Contender, Emerging Delegation), dynamic BMI computation, real-time dual inference (Classification + Regression), and native theme-adaptive diagnostic tables comparing delegation profiles against historical Olympic medal-winning medians.